

White's Conjecture and Stressed Subset Relaxations

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White proposed several conjectures concerning exchange properties of bases and the toric ideals of matroids. In this talk, I will present my thesis work on stressed subsets and the combinatorial versions of White's conjecture. Motivated by the circuit-hyperplane relaxation approach of Han, Michałek, and Weigert for sparse paving matroids, we generalize their method to stressed subset relaxations and prove that the validity of the combinatorial White's conjecture is preserved under stressed subset relaxations. We further show that the counterpart for stressed subset tightenings can be reduced to explicit low-degree cases. I will also present joint work with my advisor, Chi Ho Yuen, in which we establish the combinatorial White's conjecture for paving matroids. Finally, I will briefly mention the toric-ideal versions of White's conjecture and recent developments concerning Gröbner bases.

Keywords: Matroids, Basis exchange, White's conjecture, Relaxations.